Inference for Heterogeneous Effects using Low-Rank Estimation of Factor Slopes*

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Abstract

We study a panel data model with heterogeneous effects, allowing slopes to vary across individuals and time. To reduce dimensionality, we assume the heterogeneous slopes follow a factor structure, implying that slope matrices are estimable via low-rank regularized regression. We propose a multi-step estimation procedure incorporating sample splitting and partialing-out to enable valid inference after penalized estimation. We establish the asymptotic normality of the estimator, facilitating inference for individual-time-specific effects and their cross-sectional averages. The method's performance is evaluated through simulations and an empirical application.

Keywords: interactive effects, post-selection inference, fixed effect, panel data
JEL: C21, C23

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1 Introduction

This paper studies inference within the following panel data model:

$$y_{it} = x_{it}'\theta_{it} + \alpha_i'g_t + u_{it}, \quad i = 1, ..., N, \quad t = 1, ..., T, \tag{1.1}$$

where y_{it} is the scalar outcome of interest, x_{it} is a d-dimensional vector of observed covariates with heterogeneous slopes θ_{it}, α_i and g_t are unobserved fixed effects, and u_{it} is an unobserved error term. The model permits general heterogeneity in the sense that fixed effects appear in the model interactively and the slope θ_{it} is allowed to vary across both i and t. The main goal of this paper is to provide asymptotic inference for averages of the θ_{it} taken across subgroups in the population at specific time periods. These subgroups may consist of individual units (for inference about a specific θ_{it}), the entire cross-section, or any pre-specified subset of the cross-sectional units.

The main dimension reduction assumption employed in this paper is that θ_{it} can be expressed as

$$\theta_{it} = \lambda_i' f_t.$$

That is, we assume the slopes θ_{it} can be represented by a factor structure where λ_i is a matrix of loadings and f_t is a vector of factors. We allow f_t and g_t to have overlapping components and allow λ_i and f_t to be constant across $i \leq N$ and $t \leq T$ respectively. Thus, the model accommodates homogeneous slopes as a special case. We note that θ_{it} is not subject to rotational indeterminacy as it is a product of λ_i and f_t, so it is well-identified and cleanly estimable.

By accommodating general covariates and allowing slopes to vary across both i and t, our model differs from those in the existing low-rank estimation and interactive fixed effects literature. The factor-slope structure captures a rich spectrum of heterogeneous effects and enables hypothesis tests that are useful for understanding policy effects over the sample period. Leveraging the factor-slope structure also allows us to transform many fundamental high-dimensional inferential problems into more tractable low-dimensional problems. For example, under mild conditions, the problem of testing the hypothesis of no effect at time t—i.e., $H_0 : \theta_{it} = 0$ for all i at a given t—is equivalent to testing $f_t = 0$ almost surely. We illustrate the use of our results in such hypothesis testing problems after presenting the main results in Section 3.

It is useful to represent the model in matrix form. Let Θ_r and X_r be the $N \times T$ matrices with r^{th} component $\theta_{it,r}$ and $x_{it,r}$ respectively. Let M, Y, and U be the $N \times T$ matrices of $\alpha_i' g_t$, y_{it}, and u_{it}. Finally, let $\odot$ denote the matrix element-wise (Hadamard) product. Using this notation, the matrix form of (1.1) is

$$Y = \sum_{r=1}^{d} X_r \odot \Theta_r + M + U.$$

Under the maintained factor structure, the slope and fixed effect matrices, Θ_r and M, have rank at most equal to their associated numbers of factors. This structure motivates estimating the model parameters via low-rank estimation:

$$\begin{aligned} &\min_{\{\Theta_1,\ldots,\Theta_d,M\}} \|Y - \sum_{r=1}^{d} X_r \odot \Theta_r - M\|_F^2 + P_0(\Theta_1, ..., \Theta_d, M) \\ &\text{where} \quad P_0(\Theta_1, ..., \Theta_d, M) = \sum_{r=1}^{d} \nu_r \|\Theta_r\|_n + \nu_0 \|M\|_n \end{aligned} \tag{1.2}$$

for some tuning parameters $\nu_0, \nu_1, ..., \nu_d > 0$ and $\|.\|_F$ and $\|.\|_n$ respectively denoting the matrix Frobenius norm and the nuclear norm.

With suitable choice of tuning parameters, the low-rank estimators defined in (1.2) are consistent. However, regularization may result in large shrinkage bias in finite samples, complicating inference. This paper contributes to the literature on penalized low-rank estimation by providing an approach to obtaining valid inferential statements after applying singular value thresholding (SVT) type regularization. We use the solution of (1.2) as an initial estimate from which we extract singular vectors to serve as preliminary estimates of

λ_i and α_i. We then estimate the factors and loading iteratively via least squares. This procedure is integrated with partialing-out and sample-splitting to further alleviate the effect of regularized estimation.

Nuclear norm penalization is now a standard technique for estimating low-rank models; see, e.g., Negahban and Wainwright (2011); Recht et al. (2010); Sun and Zhang (2012); Candès and Tao (2010); Koltchinskii et al. (2011). In the matrix completion setting, Xia and Yuan (2021) and Chen et al. (2019) proposed element-wise inference of the low-rank matrix. In the econometrics literature, Bai and Ng (2019) considered iterative ridge regressions with rank constraints in pure factor models. Athey et al. (2021), Cahan et al. (2023), and Choi et al. (2024) consider the use of matrix completion methods to impute missing potential outcomes. We contribute to this literature by considering non-binary covariates with factor slopes and by providing distributional results for individual-time-specific slope coefficients and their cross-sectional averages.

Recently, Chernozhukov et al. (2023) studied inference for linear low-rank models *without* fixed effects. As it does not include fixed effects, their model is substantially simpler to estimate. Including interactive fixed effects is relevant in many empirical settings but introduces technical challenges. Specifically, estimation of the fixed effects "spills over" and has non-negligible effect on estimation of the parameters of interest. To address this complication, we introduce a partialing-out step. Relative to Chernozhukov et al. (2023), we also provide new tests about effect homogeneity that are relevant in policy studies.

Our paper is also related to the extensive panel data literature with interactive fixed effects; see, e.g., Bai (2009), Su et al. (2015), Ahn et al. (2013) and Breitung and Hansen (2021). Moon and Weidner (2018) use nuclear-norm regularization for estimating interactive fixed effects in a model with homogeneous slopes. Studying slope heterogeneity has also been an important topic for panel data. For example, Chamberlain and Hirano (1999) and Pesaran (2006) consider models with coefficients that are not time-varying but are heterogeneous across individuals. Bonhomme and Manresa (2015) and Su et al. (2016) study settings where coefficients are assumed to be homogeneous within latent groups and estimation proceeds by simultaneously estimating group-specific effects. To allow for heterogeneity in both i and t, Feng et al. (2017) assume that slopes are functions of observed time varying categorical variables, and Su and Wang (2017) considers a model in which slope coefficients vary smoothly over time so are locally time invariant. Horváth and Trapani (2016) studied a dynamic model where the coefficient can be decomposed into a common mean and an iid sequence across time. In other recent work, Keane and Neal (2020) and Lu and Su (2023) consider models where heterogeneous slopes follow an additive two-way fixed effects structure, and we note that one could use our framework to develop tests for whether low rank slope coefficient admits a two-way additive structure. One could also adopt hierarchical Bayesian inference for the heterogeneous coefficients by assuming they are drawn from hierarchical priors as in, e.g., Hsiao et al. (1999). Relative to these approaches, we use a different dimension reduction strategy that allows variation in coefficients across individuals and over time without relying on additional structure over the factors or factor loadings. Within this structure, we are able to provide inference for a variety of effects.

As an empirical illustration, we apply the proposed heterogeneous model with factor slopes to estimating the effect of minimum wage on employment. We find that the estimated effect is neither uniformly positive nor uniformly negative, but is instead heterogeneous across both the time and cross-sectional dimension.

The rest of the paper is organized as follows. Section 2 introduces the post-SVT algorithms that define our estimators. Section 3 provides asymptotic theory. Section 4 presents simulation results. Section 5 applies the proposed models to the minimum wage example. We provide proofs in a supplementary appendix.

2 The Model

We consider the model

$$
\begin{aligned}
y_{it} &= \theta_{it} x_{it} + \alpha_i' g_t + u_{it}, \quad i = 1, ..., N, \quad t = 1, ..., T, \text{ with} \\
\theta_{it} &= \lambda_i' f_t, \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (2.1) \\
x_{it} &= \mu_{it} + e_{it}.
\end{aligned}
$$

We observe (y_{it}, x_{it}), and the goal is to conduct inference on θ_{it} or on averages of θ_{it} across groups of cross-sectional units. The term $\alpha_i' g_t$ represents interactive fixed effects, as in Bai (2009). For ease of presentation, we focus on the case where x_{it} is univariate, i.e., $\dim(x_{it}) = 1$. The extension to the multivariate case is straightforward and is provided in Appendix A in the supplementary material.

The third equation in (2.1) gives the decomposition of x_{it}, which contains a mean structure μ_{it} as well as the noise e_{it}. As we shall explain later, our estimation strategy will be based on the well known "residual regression" from Robinson (1988) that essentially uses the estimated residual e_{it} as regressors. Thus, we will require a model for x_{it} in which μ_{it} can be estimated relatively well. To achieve this goal, we assume a factor model for x_{it} in our formal analysis such that

$$\mu_{it} = l_i' w_t \tag{2.2}$$

with loadings l_i and factors w_t. We note that many other structures for x_{it} could also be accommodated with appropriate adaptation of formal conditions.

We assume that $\{\lambda_i, \alpha_i, l_i\}$ are deterministic sequences, while $\{f_t, g_t, w_t\}$ are random. We allow for weak dependence among $\{f_t : t \leq T\}$. We also allow a special case where factors are constant. Considering constant factors allows us to accommodate homogeneous models where $\theta_{it} = \theta$ for all (i, t), which can be obtained by setting $\lambda_i = \lambda$ and $f_t = f$ for all (i, t).

Throughout, we assume the ranks $\dim(\lambda_i) = K_1$ and $\dim(\alpha_i) = K_2$ are fixed. We note that this differs from much of the matrix completion literature which explicitly considers model sequences where ranks may increase with the sample size. We leave the extension to the case where ranks are allowed to increase to future work. We also take the ranks to be known in the formal development, but we do present an approach to estimating the ranks in Section 2.4, and we employ this method to estimate the ranks in the simulation study. Under the assumption that factors are strong (which we maintain in our formal analysis), the provided algorithm consistently estimates the ranks and estimation of the ranks does not impact the remaining asymptotic results.

2.1 Nuclear Norm Penalized Estimation

Let (Y, X, U) be $N \times T$ matrices of (y_{it}, x_{it}, u_{it}). Then (2.1) may be expressed in matrix form as

$$Y = M + X \odot \Theta + U$$

where Θ and M are matrices of $(\theta_{it}, \alpha_i' g_t)$.[1]

Motivated by the low-rank structure of M and Θ, we start with the following penalized nuclear-norm optimization problem:

$$\begin{aligned} (\widetilde{\Theta}, \widetilde{M}) &= \arg\min_{\Theta, M} F(\Theta, M), \\ F(\Theta, M) &:= \|Y - M - X \odot \Theta\|_F^2 + \nu_0 \|M\|_n + \nu_1 \|\Theta\|_n \end{aligned} \tag{2.3}$$

for some tuning parameters $\nu_0, \nu_1 > 0$. We note that this minimization problem is convex, so readily solved. In the following, we sketch one algorithm for obtaining the solution.

For a fixed matrix Y, let $U_Y D_Y V_Y' = Y$ be its singular value decomposition. Define the

[1] Model (2.1) can be rewritten in the form of a trace regression model studied by Negahban and Wainwright (2011): $Y_j = \operatorname{trace}(\mathfrak{X}_j' \Gamma) + \varepsilon_j$, where $\mathfrak{X}_j' = (\iota_t e_i', \ x_{it} \iota_t e_i')$ and Γ stacks the matrices M and Θ. However, applying trace regression methods alone—treating Γ as a low-rank matrix—is not sufficient to enable inference on its individual elements. While additional debiasing steps could, in principle, address this issue, the extreme sparsity of $\mathfrak{X}_j$ poses significant challenges for standard debiasing techniques. For related discussion in the simpler setting of matrix completion, see Xia and Yuan (2021) and Chen et al. (2019).

singular value thresholding (SVT) operator

$$S_\lambda(Y) = U_Y D_\lambda V_Y',$$

where D_λ is defined by replacing the diagonal entry D_{ii} of D by $\max\{D_{ii} - \lambda, 0\}$. That is, $S_\lambda(Y)$ applies soft-thresholding on the singular values of Y.

The solution of (2.3) can be obtained by iteratively using SVT. Given Θ, solving for M in (2.3) leads to the solution

$$S_{\nu_0/2}(Z_\Theta) = \arg\min_M \|Z_\Theta - M\|_F^2 + \nu_0\|M\|_n$$

for $Z_\Theta = Y - X \odot \Theta$. Similarly, given M, let $Z_M = Y - M$. We solve for Θ via

$$\Theta_M := \arg\min_\Theta \|Z_M - X \odot \Theta\|_F^2 + \nu_1\|\Theta\|_n,$$

which satisfies the following KKT condition (Ma et al., 2011): For any $\tau > 0$,

$$\Theta_M = S_{\tau\nu_1/2}(\Theta_M - \tau X \odot (X \odot \Theta_M - Z_M)).$$

As such, we employ the following algorithm to iteratively solve for $\widetilde{M}$ and $\widetilde{\Theta}$ as the global solution to (2.3).

Algorithm 2.1. Compute the nuclear-norm penalized regression as follows:

Step 1: Fix the "step size" $\tau \in (0, 1/\max_{it} x_{it}^2)$. Initialize Θ_0, M_0 and set $k = 0$.

Step 2: Let

$$\begin{aligned} \Theta_{k+1} &= S_{\tau\nu_1/2}(\Theta_k - \tau X \odot (X \odot \Theta_k - Y + M_k)), \\ M_{k+1} &= S_{\nu_0/2}(Y - X \odot \Theta_{k+1}). \end{aligned}$$

Set k to $k + 1$.

Step 3: Repeat step 2 until convergence.

In practice, we set $\tau = (1-\epsilon)/\max_{it} x_{it}^2$ for some small $\epsilon > 0$.

We note that, given M, updating Θ_M is a standard gradient descent procedure (e.g., Beck and Teboulle (2009)), and therefore standard convergence analysis for the objective function $F(\Theta, M)$ applies. In particular, Proposition 2.1 verifies that the evaluated objective function is monotonically decreasing and the algorithm converges from an arbitrary initial value.

Proposition 2.1. *Let $(\widetilde{\Theta}, \widetilde{M})$ be a global minimum for $F(\Theta, M)$. Then for any $\tau \in (0, 1/\max_{it} x_{it}^2)$, and any initial Θ_0, M_0, we have:*

$$F(\Theta_{k+1}, M_{k+1}) \leq F(\Theta_{k+1}, M_k) \leq F(\Theta_k, M_k),$$

for each $k \geq 0$. In addition, for all $k \geq 1$,

$$F(\Theta_{k+1}, M_{k+1}) - F(\widetilde{\Theta}, \widetilde{M}) \leq \frac{1}{k\tau} \|\Theta_1 - \widetilde{\Theta}\|_F^2. \tag{2.4}$$

2.2 Intuition for Main Algorithm

Nuclear-norm penalized estimators are generally inappropriate for inference because they may suffer from substantial shrinkage bias. To address this, we consider a debiased estimator. In this section, we provide a heuristic argument that highlights the chief complications that arise in obtaining reliable asymptotic approximations in our setting. We explain how partialing-out, sample-splitting, and iterative OLS estimation are used to address these issues. Part of the argument deals with the rotation of the factors, which, to the best of our knowledge, has not appeared previously in the post-regularization inference literature.

To illustrate the key issues, we work in a simplified setting where $\dim(f_t) = \dim(g_t) = 1$. Suppose we have preliminary estimates $\widetilde{\lambda}_i$ and $\widetilde{\alpha}_i$ obtained by extracting the first singular vector from $\widetilde{\Theta}$ and $\widetilde{M}$ obtained from (2.3). One might then attempt to estimate f_t as the coefficient on $\widetilde{\lambda}_i x_{it}$ from a regression of y_{it} on $\widetilde{\lambda}_i x_{it}$ and $\widetilde{\alpha}_i$. Letting $\check{f}_t$ denote this coefficient,

$\check{f}_t$ would have the following expansion: For some $\widehat{Q}$ and rotation matrices H_1 and H_2,

$$\begin{aligned}
\sqrt{N}(\check{f}_t - H_1^{-1} f_t) &= \widehat{Q}\frac{1}{\sqrt{N}}\sum_{i=1}^N \lambda_i x_{it} u_{it} + \widehat{Q}\widetilde{\Delta}_\alpha + \widehat{Q}\widetilde{\Delta}_\lambda + o_P(1) \\
\widetilde{\Delta}_\alpha &= \frac{1}{\sqrt{N}}\sum_{i=1}^N \lambda_i x_{it}(\widetilde{\alpha}_i - H_2\alpha_i) \\
\widetilde{\Delta}_\lambda &= \frac{1}{\sqrt{N}}\sum_{i=1}^N \lambda_i x_{it}^2(\widetilde{\lambda}_i - H_1\lambda_i) \qquad (2.5)
\end{aligned}$$

Terms $\widetilde{\Delta}_\alpha$ and $\widetilde{\Delta}_\lambda$ capture the effect of estimation error in $\widetilde{\alpha}$ and $\widetilde{\lambda}$, and generally do not vanish asymptotically. In the presence of these terms, regularization bias results in a failure of asymptotic normality.

2.2.1 Partialing-Out and Sample Splitting

We deal with terms $\widetilde{\Delta}_\alpha$ and $\widetilde{\Delta}_\lambda$ by leveraging additional structure on the observed variable x_{it} and using partialing-out and sample splitting. Write

$$x_{it} = \mu_{it} + e_{it}, \qquad (2.6)$$

where μ_{it} is the mean process of x_{it} which is assumed to capture both time series dependence and strong sources of cross-sectional correlation. Specifically, we will maintain the assumption that e_{it} is a zero-mean process that is serially independent and cross-sectionally weakly dependent throughout our formal analysis.

Using this decomposition of x_{it}, we can produce a "partialed-out" version of the model:

$$\dot{y}_{it} = \alpha_i' g_t + e_{it}\lambda_i' f_t + u_{it}, \qquad \text{where } \dot{y}_{it} = y_{it} - \mu_{it}\theta_{it}. \qquad (2.7)$$

The transformed model now contains partialed-out regressors e_{it}. Using these regressors, we can estimate f_t by regressing the estimated $\dot{y}_{it}$ onto $(\widetilde{\alpha}_i, \widetilde{\lambda}_i e_{it})$. The resulting estimator of f_t

then admits the expansion

$$
\begin{aligned}
\sqrt{N}(\widehat{f}_t - H_1^{-1} f_t) &= \widehat{Q}\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\lambda_i e_{it}u_{it} + \Delta_\alpha g_t + \Delta_\lambda f_t + o_P(1), \\
\text{where} \quad \Delta_\alpha &:= \widehat{Q}\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\lambda_i e_{it}(\widetilde{\alpha}_i - H_2\alpha_i)', \\
\Delta_\lambda &:= \widehat{Q}\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\lambda_i(\mathsf{E}e_{it}^2)(\widetilde{\lambda}_i - H_1\lambda_i)'.
\end{aligned} \tag{2.8}
$$

It is immediate that $\Delta_\alpha = o_P(1)$ if e_{it} is mean zero, independent of $\lambda_i(\widetilde{\alpha}_i - H_2\alpha_i)$, and sufficient moments exist. The plausibility of these conditions relies on partialing-out the strongly dependent components in x_{it} and is tightly tied to the so-called *Neyman orthogonality* that has been shown to be important in obtaining valid inference after high-dimensional estimation in a variety of contexts; see, e.g., Chernozhukov et al. (2018).

Implementing partialing-out in (2.7) requires a high-quality estimate of e_{it}, which in turn requires some structure on μ_{it}. In our formal analysis, we assume $\mu_{it} = l_i' w_t$ and assume that $\dim(w_t)$ is fixed, with (l_i, w_t) representing loadings and factors. This structure allows for strong intertemporal and cross-sectional dependence in x_{it}. We permit overlap between w_t and (f_t, g_t) but do not require it. In cases where perfect overlap occurs such that all components are driven by the same small set of common factors, the common factor approach of Pesaran (2006) could serve as an alternative method to estimate the factors jointly, though we do not pursue that direction in this paper.

As noted previously, other structures of μ_{it} could also be imposed with all results going through as long as μ_{it} is sufficiently well-estimable. For instance, it is straightforward to obtain high quality estimates with the simple "many means" model:

$$
x_{it} = \mu_i + e_{it}, \quad \mu_i = \mathsf{E}x_{it}.
$$

To illustrate how the analysis of our model can be extended to accommodate other structures on x_{it}, we present results for estimation of θ_{it} when x_{it} follows the many means model in Appendix H in the supplement. The bottom line is that as long as we can obtain a sufficiently accurate estimator of e_{it}, the asymptotic distribution of our estimator of the target parameter is equivalent to that obtained using the true e_{it}. That is, the impact of estimating μ_{it} is asymptotically negligible. This intuition is the same as that of the residual regression literature for post-selection inference and double machine learning inference; see, e.g., Robinson (1988); Belloni et al. (2014).

To complete the argument that $\Delta_\alpha = o_P(1)$, we also use sample-splitting. For a fixed t, let $I \subset \{1, ..., T\}\backslash t$ be a set of time indexes, and let

$$D_I = \{(y_{is}, x_{is}) : i \leq N, s \in I\}.$$

Rather than using the full sample to obtain initial estimates of $\widetilde{\lambda}_i$ and $\widetilde{\alpha}_i$, we run the nuclear-norm optimization using only data D_I. Because $t \notin I$ and e_{it} is assumed to be serially independent, $\widetilde{\lambda}_i$ and $\widetilde{\alpha}_i$ are independent of e_{it}. This independence then allows us to easily verify that Δ_α and similar terms vanish asymptotically.

2.3 Formal Estimation Algorithm.

We now state the full estimation algorithm for θ_{it} for some fixed t. The algorithm is stated in the case where $x_{it} = l_i' w_t + e_{it}$ with $\mathsf{E} e_{it} = 0$.

For a fixed t, a time point of interest, we randomly split the sample into $\{1, ..., T\}/\{t\} = I_t \cup I_t^c$, where (I_t, I_t^c) depends on t. For notational simplicity, we suppress the dependence on t and use (I, I^c) hereafter.

Algorithm 2.2. Estimate θ_{it} as follows.

Step 1. Estimate the structure $x_{it} = \mu_{it} + e_{it}$. Use the principal components (PC) estimator to obtain $(\widehat{\mu}_{it}, \widehat{e}_{it})$ for all $i = 1, ..., N$, $t = 1, ..., T$.

Step 2: Sample splitting. Randomly split the sample into $\{1, ..., T\}/\{t\} = I \cup I^c$, so that

$|I|_0 = [(T-1)/2]$. Denote the $N \times |I|_0$ matrices of (y_{is}, x_{is}) for observations $s \in I$ by Y_I, X_I. Run nuclear-norm penalized regression:

$$(\widetilde{M}_I, \widetilde{\Theta}_I) := \arg\min_{M,\Theta} \|Y_I - M - X_I \odot \Theta\|_F^2 + \nu_0 \|M\|_n + \nu_1 \|\Theta\|_n. \tag{2.9}$$

Let $\widetilde{\Lambda}_I = (\widetilde{\lambda}_1, ..., \widetilde{\lambda}_N)'$ be the $N \times K_1$ matrix whose columns are defined as $\sqrt{N}$ times the first K_1 eigenvectors of $\widetilde{\Theta}_I \widetilde{\Theta}_I'$. Let $\widetilde{A}_I = (\widetilde{\alpha}_1, ..., \widetilde{\alpha}_N)'$ be the $N \times K_2$ matrix whose columns are defined as $\sqrt{N}$ times the first K_2 eigenvectors of $\widetilde{M}_I \widetilde{M}_I'$.

Step 3. Estimate components for "partialing-out." Using $\widetilde{A}_I$ and $\widetilde{\Lambda}_I$, obtain

$$(\widetilde{f}_s, \widetilde{g}_s) := \arg\min_{f_s, g_s} \sum_{i=1}^{N} (y_{is} - \widetilde{\alpha}_i' g_s - x_{is} \widetilde{\lambda}_i' f_s)^2, \quad s \in I^c \cup \{t\}.$$

Update estimates of loadings as

$$(\dot{\lambda}_i, \dot{\alpha}_i) = \arg\min_{\lambda_i, \alpha_i} \sum_{s \in I^c \cup \{t\}} (y_{is} - \alpha_i' \widetilde{g}_s - x_{is} \lambda_i' \widetilde{f}_s)^2, \quad i = 1, ..., N.$$

Step 4. Estimate (f_t, λ_i) for use in inference about θ_{it}. Define $\widehat{y}_{is} = y_{is} - \widehat{\mu}_{is} \dot{\lambda}_i' \widetilde{f}_s$ and $\widehat{e}_{is} = x_{is} - \widehat{\mu}_{is}$. Let

$$\begin{aligned}
(\widehat{f}_{I,s}, \widehat{g}_{I,s}) &:= \arg\min_{f_s, g_s} \sum_{i=1}^{N} (\widehat{y}_{is} - \widetilde{\alpha}_i' g_s - \widehat{e}_{is} \widetilde{\lambda}_i' f_s)^2, \quad s \in I^c \cup \{t\} \\
(\widehat{\lambda}_{I,i}, \widehat{\alpha}_{I,i}) &:= \arg\min_{\lambda_i, \alpha_i} \sum_{s \in I^c \cup \{t\}} (\widehat{y}_{is} - \alpha_i' \widehat{g}_{I,s} - \widehat{e}_{is} \lambda_i' \widehat{f}_{I,s})^2, \quad i = 1, ..., N.
\end{aligned}$$

Step 5. Exchange I and I^c. Repeat steps 2-4 with I and I^c exchanged to obtain $(\widehat{\lambda}_{I^c,i}, \widehat{f}_{I^c,s} : s \in I \cup \{t\}, i \leq N)$.

Step 6. Estimate θ_{it}. Obtain the estimator of θ_{it}:

$$\widehat{\theta}_{it} := \frac{1}{2} [\widehat{\lambda}_{I,i}' \widehat{f}_{I,t} + \widehat{\lambda}_{I^c,i}' \widehat{f}_{I^c,t}].$$

Remark 2.1. We split the sample as $\{1,\ldots,T\} = I \cup I^c \cup \{t\}$ to allow weak dependence across individuals that is not captured by the latent factors. Allowing this dependence is important because low-rank models require the number of factors to be relatively small, limiting their ability to capture all forms of cross-sectional dependence. Splitting along the time index t also enables estimation of the full set of loading parameters $\{\lambda_i, \alpha_i\}_{i\leq N}$, which is useful for analyzing average effects across individuals.

A limitation of splitting over t is that it prevents us from exploiting the natural temporal ordering of the data and requires the noise e_{it} to be serially independent. As a result, this approach cannot accommodate dynamic panel models and imposes strong assumptions on temporal dependence. In many applications, it may also be of interest to estimate the full set of factor parameters $\{f_t, g_t\}_{t\leq T}$. In such cases, one can instead split across individuals. Doing so would allow for weak serial dependence in e_{it}, provided that the error terms are cross-sectionally independent.

We note that, in this paper, we chiefly use sample splitting as a technical device to facilitate the theoretical proof that the nuclear-norm penalized estimator does not influence the asymptotic distribution of the final estimator. In simpler low-rank matrix completion settings without fixed effects, recent work has shown how to establish asymptotic results without sample splitting (e.g., Choi et al. (2024); Chen et al. (2019)). Extending these arguments to models with interactive fixed effects may be possible, though doing so appears technically challenging at a minimum.

2.4 Choosing the tuning parameters

We adopt a simple plug-in approach to choosing the tuning parameters ν_1 and ν_0. The "scores" of the penalized regression are given by $2U$ and $2X \odot U$. We then wish to choose tuning parameters (ν_0, ν_1) such that we achieve score domination in the sense that

$$2\|U\| < (1-c)\nu_0, \quad 2\|X \odot U\| < (1-c)\nu_1 \tag{2.10}$$

for some $c > 0$ with high probability where $\|\cdot\|$ denotes the matrix operator norm.

First, we pre-specify $\widetilde{\nu}_0, \widetilde{\nu}_1 \asymp \sqrt{T+N}$, which satisfy the rate requirement of $2\|U\| = O_P(\widetilde{\nu}_0)$ and $2\|X \odot U\| = O_P(\widetilde{\nu}_1)$. We then apply Algorithm 2.2 without sample splitting or partialing-out. That is, we conduct the nuclear-norm penalized estimation and least squares steps (Step 2 and Step 3) using the full dataset. Let $\widetilde{u}_{it} = y_{it} - \alpha_i'\widetilde{g}_t - x_{it}\lambda_i'\widetilde{f}_t$ be the residuals after Step 3 (on the full data). Without sample-splitting or partialing-out, these two steps do not deliver an estimator ready for inference. However, the estimated residuals are consistent, which is sufficient for obtaining a consistent estimator for $\mathsf{Var}(u_{it})$: $\widetilde{\sigma}^2_{u,i} = \frac{1}{T}\sum_t \widetilde{u}^2_{it}$.

We then update ν_0, ν_1 as follows. Let Z be an $N \times T$ matrix whose elements are independently generated from $\mathcal{N}(0, \widetilde{\sigma}_{u,i})$. Let $Q(W; m)$ denote the m^{th} quantile of random variable W. Set

$$\nu_0 = 2(1+c_1)Q(\|Z\|; 1-\delta_{NT}), \quad \nu_1 = 2(1+c_1)Q(\|X \odot Z\|; 1-\delta_{NT}),$$

which respectively denote $2(1+c_1)$ times the $1-\delta_{NT}$ quantile of $\|Z\|$ and $\|X \odot Z\|$. These quantiles can be computed via simulation. Using these values, (2.10) holds with probability $1-\delta_{NT}$. Our theoretical results make use the condition that $\delta_{NT} \to 0$. In practice, we simply set δ_{NT} to a small number. For example, we set $c_1 = 0.1$ and $\delta_{NT} = 0.05$ in our simulation and empirical examples.

The ranks of the low-rank matrices, $\mathsf{rank}(\Theta_0) = \dim(f_t)$ and $\mathsf{rank}(M_0) = \dim(g_t)$, must also be set in practice. In our formal results, we assume these ranks are known. In practice, they can be estimated by counting the number of large singular values in estimates of the matrix parameters obtained in a first step nuclear-norm penalized regression step. Specifically, we run the nuclear-norm penalized regression

$$(\widetilde{M}, \widetilde{\Theta}) = \arg\min_{M,\Theta} \|Y - M - X \odot \Theta\|_F^2 + \nu_1\|M\|_n + \nu_2\|\Theta\|_n \tag{2.11}$$

in the full dataset and without sample splitting. We then estimate the ranks as

$$\widehat{K}_1 = \sum_i 1\{\psi_i(\widetilde{M}) \geq 0.1(\nu_1 \|\widetilde{M}\|)^{1/2}\}, \quad \widehat{K}_2 = \sum_i 1\{\psi_i(\widetilde{\Theta}) \geq 0.1(\nu_2 \|\widetilde{\Theta}\|)^{1/2}\}$$

where $\psi_i(A)$ returns the i^{th} largest singular-value of matrix A. Proposition E.1 in the appendix verifies that these estimated ranks are consistent:

$$P(\widehat{K}_1 = K_1, \widehat{K}_2 = K_2) \to 1.$$

Consistent rank estimation implies that estimating ranks has no asymptotic impact for inference.

3 Asymptotic Results

In this section, we present our main formal results. We start by defining inferential targets in Section 3.1. We then present assumptions and our main theoretical results.

3.1 Parameters of interest relevant to policy studies

Our main inferential theory establishes the rate of convergence and asymptotic normality for a *group average effect*. Fix a cross-sectional subgroup

$$\mathcal{G} \subseteq \{1, 2, ..., N\}.$$

We are interested in inference for the group average effect at a fixed $t \leq T$:

$$\bar{\theta}_{\mathcal{G},t} := \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \theta_{it},$$

where $|\mathcal{G}|_0$ denotes the group size. The group size can be either fixed or grow with N. This structure admits two interesting cases as extremes: (i) $\mathcal{G} = \{i\}$ for any fixed i, which allows inference for a fixed individual; and (ii) $\mathcal{G} = \{1, 2, ..., N\}$, which allows inference for the cross-sectional average effect $\bar{\theta}_t := \frac{1}{N}\sum_{i=1}^{N}\theta_{it}$.

The group average effect provides answers to various questions related to policy studies, e.g., the effect of the minimum wage in a state of interest or on average in the country at different points in time. Inference about $\bar{\theta}_{\mathcal{G},t}$ is also relevant to answering many questions that are important for understanding effects of policies over the sample period. For example, one may be interested in the following hypotheses related to policy effects during the sample period:

- **Tests of homogeneity across groups in a given time period.** Given a finite set of groups—$\mathcal{G}_1, ..., \mathcal{G}_J$—one may wish to test the hypothesis of homogeneous average group effects at a given point in time:

  $$H_0^1 : \bar{\theta}_{\mathcal{G}_1,t} = ... = \bar{\theta}_{\mathcal{G}_J,t}.$$

  For instance, we may be interested in asking whether the average effect of the minimum wage in states in two different regions of the country, such as historically poorer and historically wealthier regions, are the same.

- **Test of joint significance in a given time period.** One might be interested in testing that all effects in a given time period are zero:

  $$H_0^2 : \theta_{it} = 0 \text{ for all } i.$$

- **Tests of constant group effect across two periods.** For two time points t_1 and t_2 and a fixed group $\mathcal{G}$, one may be interested in testing the hypothesis that average group effects are constant:

$$H_0^3 : \bar{\theta}_{\mathcal{G},t_1} = \bar{\theta}_{\mathcal{G},t_2}.$$

For instance, t_1 and t_2 can be respectively the period before and after a large minimum wage change.

- **Test of within-group homogeneity.**

  Fix a group $\mathcal{G}$. Consider testing:

$$H_0^4 : \theta_{1,t} = ... = \theta_{N_0,t}, \quad \forall t \leq T.$$

  For instance, one may be interested in testing whether the minimum wage effect is the same within the group of high (or low) minimum wage states.

We show how valid tests of these hypotheses result as simple extensions of our main results in Section 3.4. Specifically, we examine tests of the first two hypotheses in that section. We leave the analysis of tests of the other two hypotheses to the supplementary appendix.

3.2 Assumptions

To establish our formal results, we rely on sufficient conditions outlined in this section.

Our first assumption places restrictions on the latent factors. These conditions will hold for serially independent sequences.

Assumption 3.1. *As $T \to \infty$, the sub-samples (I, I^c) satisfy:*

$$\frac{1}{|I|_0} \sum_{t \in I} f_t f_t' = \frac{1}{T} \sum_{t=1}^{T} f_t f_t' + O_P(T^{-1/2}) = \frac{1}{|I^c|_0} \sum_{t \in I^c} f_t f_t',$$
$$\frac{1}{|I|_0} \sum_{t \in I} g_t g_t' = \frac{1}{T} \sum_{t=1}^{T} g_t g_t' + O_P(T^{-1/2}) = \frac{1}{|I^c|_0} \sum_{t \in I^c} g_t g_t'.$$

In addition, there is a $c > 0$ such that all the eigenvalues of $\frac{1}{T}\sum_{t=1}^{T} f_t f_t'$ and $\frac{1}{T}\sum_{t=1}^{T} g_t g_t'$ are bounded from below by c almost surely.

The next assumption imposes that the factors are strong. In addition, because our algorithm requires constructing eigenvectors in order to learn (λ_i, α_i), we require the matrices from which we extract these eigenvectors to have distinct eigenvalues as this allows identification of the corresponding eigenvectors. These conditions are strong, though standard in the factor modeling literature. It may be interesting, but is beyond the scope of the present work, to consider estimation and inference in the presence of weak factors.

Assumption 3.2 (Valid factor structures with strong factors). *There are constants $c_1 > ... > c_{K_1} > c > 0$, and $c_1' > ... > c_{K_2}' > c > 0$, so that up to a term $o_P(1)$,*

(i) c_j' equals the j^{th} largest eigenvalue of $(\frac{1}{T}\sum_t g_t g_t')^{1/2}\frac{1}{N}\sum_{i=1}^{N}\alpha_i\alpha_i'(\frac{1}{T}\sum_t g_t g_t')^{1/2}$ for all $j = 1, ..., K_1$, and

(ii) c_j equals the j^{th} largest eigenvalue of $(\frac{1}{T}\sum_t f_t f_t')^{1/2}\frac{1}{N}\sum_{i=1}^{N}\lambda_i\lambda_i'(\frac{1}{T}\sum_t f_t f_t')^{1/2}$ for all $j = 1, ..., K_2$.

Proposition E.1 in the supplementary appendix shows that the nuclear-norm regularized matrix estimators defined in (2.3) are consistent under the Frobenius norm and provides the rate of convergence under Assumptions 3.1 and 3.2. It extends known results from the low-rank estimation literature to the multi-dimensional case.

To obtain the asymptotic distribution of our estimator, we make a number of additional assumptions. Throughout, let (F, G, W, E, U) be the $T \times K$ matrices of $(f_t, g_t, w_t, e_{it}, u_{it})$, where K differs for different quantities.

Assumption 3.3 (Dependence). *(i) $\{e_{it}, u_{it}\}$ are independent across t; $\{e_{it}\}$ are also conditionally independent across t given $\{F, G, W, U\}$; $\{u_{it}\}$ are also conditionally independent across t given $\{F, G, W, E\}$.*

(ii) $\mathsf{E}(e_{it}|u_{it}, w_t, g_t, f_t) = 0, \quad \mathsf{E}(u_{it}|e_{it}, w_t, g_t, f_t) = 0$. *Also, $\mathsf{E}e_{it}^2$ does not vary across t.*

(iii) Let $\Omega_{NT} = X \odot U = (\kappa_1, ..., \kappa_T)$ be an $N \times T$ matrix . The columns $\{\kappa_t\}_{t \leq T}$ are independent sub-gaussian random vectors with $\mathsf{E}\kappa_t = 0$. More specifically, there is $C > 0$ such that

$$\max_{t \leq T} \sup_{\|x\|=1} \mathsf{E} \exp(s\kappa_t' x) \leq \exp(s^2 C), \quad \forall s \in \mathbb{R}.$$

(iv) Weak conditional cross-sectional dependence: Let $\mathcal{W} = (F, G, W)$. Let $\omega_{it} = u_{it}e_{it}$, and let c_i be a bounded nonrandom sequence. Almost surely,

$$\max_{t \leq T} \frac{1}{N^3} \sum_{i,j,k,l \leq N} |\mathsf{Cov}(e_{it}e_{jt}, e_{kt}e_{lt}|\mathcal{W}, U)| < C, \quad \max_{t \leq T} \|\mathsf{E}(u_t u_t'|\mathcal{W}, E)\| < C$$
$$\max_{t \leq T} \mathsf{E}(|\frac{1}{\sqrt{N}} \sum_{i=1}^N c_i \omega_{it}|^4|\mathcal{W}) < C, \quad \max_{t \leq T} \mathsf{E}(|\frac{1}{\sqrt{N}} \sum_{i=1}^N c_i e_{it}|^4|\mathcal{W}, U) < C$$
$$\max_{t \leq T} \mathsf{E}(|\frac{1}{\sqrt{N}} \sum_{i=1}^N c_i u_{it}|^4|\mathcal{W}, E) < C, \quad \max_{t \leq T} \max_{i \leq N} \frac{1}{N} \sum_{k,j \leq N} |\mathsf{E}(e_{kt}e_{it}e_{jt}|\mathcal{W}, U)| < C$$
$$\max_{t \leq T} \max_{i \leq N} \sum_{j=1}^N |\mathsf{Cov}(e_{it}^m, e_{jt}^r|\mathcal{W}, U)| < C, \quad m, r \in \{1, 2\}$$
$$\max_{t \leq T} \max_{i \leq N} \frac{1}{N} \sum_{k,j \leq N} |\mathsf{Cov}(\omega_{it}\omega_{jt}, \omega_{it}\omega_{kt}|\mathcal{W})| < C, \quad \max_{t \leq T} \max_{i \leq N} \sum_{j=1}^N |\mathsf{Cov}(\omega_{it}, \omega_{jt}|\mathcal{W})| < C.$$

Assumption 3.3 (i) imposes independence across t. This condition is strong and typically not needed in pure factor models or linear panel data models. We impose the independence condition as we are splitting the sample over time. We focus on the case where we consider sample splitting along t as it allows estimation of all the loading parameters $\{\lambda_i, \alpha_i\}_{i \leq N}$. Having these loadings is useful for analyzing average effects across individuals.

Condition (ii) requires $\mathsf{E}(e_{it}|u_{it}, w_t, g_t, f_t) = 0$. Recall that $x_{it} = l_i' w_t + e_{it}$, which means the residual term in x_{it} is mean independent of all factors and the residual in y_{it} and thus x_{it} should be exogenous.

Before turning to the rest of the assumptions, we note an alternative would be to consider sample splitting along the i dimension. In this case, all results immediately carry over by simply exchanging the role of i and t in all estimation steps and technical arguments. We note that this alternative may be preferable in empirical settings where independence across individuals after removing the factor structure seems more reasonable than assuming that the factor structure accounts for all sources of temporal dependence.

Assumption 3.3 (ii) imposes mean independence restrictions that serve to identify our parameters of interest and also imposes that variances of residuals are time invariant.

Assumption 3.3 (iii) requires sub-Gaussianity of the distribution of $x_{it}u_{it}$, which enables application of certain concentration inequalities.

Assumption 3.3 (iv) imposes restrictions on cross-sectional dependence but does allow for weak cross-sectional dependence among the idiosyncratic errors. The conditions in this assumption are high-level but can readily be verified under a variety of lower level conditions.

For example, consider the case where individuals can be classified into many non-overlapping "blocks"—$B_1, ..., B_n$ where $n \asymp N$—such that there are finitely many elements within each block and (u_{it}, e_{it}) are arbitrarily dependent within blocks but independent across blocks. To verify $\max_{t \leq T} \frac{1}{N^3} \sum_{i,j,k,l \leq N} |\mathsf{Cov}(e_{it}e_{jt}, e_{kt}e_{lt}|\mathcal{W}, U)| < C$ under this structure, we let $B(i)$ denote the unique block to which individual i belongs. Then write $A(ij, kl) := |\mathsf{Cov}(e_{it}e_{jt}, e_{kt}e_{lt}|\mathcal{W}, U)|$. Note that $A(ij, kl) = 0$ when $k, l \notin B(i) \cup B(j)$; $j, k, l \notin B(i)$; or several other analogous cases. Further, suppose $\mathbb{E}(e_{it}^4|\mathcal{W}, U) < C$ almost surely. Then

$$\begin{aligned} \frac{1}{N^3} \sum_{i,j,k,l \leq N} A(ij, kl) &= \frac{1}{N^3} \sum_{i,j} \sum_{k,l \in B(i) \cup B(j)} A(ij, kl) + \frac{1}{N^3} \sum_{i} \sum_{j \notin B(i)} \sum_{k \notin B(i)} \sum_{l \in B(i)} A(ij, kl) \\ &\leq CN^{-1} + C. \end{aligned}$$

The inequality follows because the block size $B(i)$ is finite uniformly in i. The remaining conditions in Assumption 3.3 (iv) can be verified similarly.

We next make a high level assumption that two central limit theorems (CLTs) hold. The assumption is similar to conditions imposed in other factor model and matrix completion settings; see, e.g., Cahan et al. (2023).

Assumption 3.4 (CLT). *As $N, T \to \infty$,*

$$\begin{pmatrix} V_{\lambda 2}^{-1/2} \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \lambda_i e_{it} u_{it}, \\ V_{f,\mathcal{G}}^{-1/2} \frac{1}{\sqrt{|\mathcal{G}|_0 T}} \sum_{i \in \mathcal{G}} \sum_{s=1}^{T} \Omega_i f_s e_{is} u_{is} \end{pmatrix} \to^d \mathcal{N}(0, I)$$

where for $\Omega_i = (\frac{1}{T} \sum_{s=1}^{T} f_s f_s' \mathsf{E} e_{is}^2)^{-1}$,

$$V_{\lambda 2} = \mathsf{Var}\left(\frac{1}{\sqrt{N}} \sum_{i=1}^{N} \lambda_i e_{it} u_{it} \middle| F \right) \quad V_{f,\mathcal{G}} = \frac{1}{T} \sum_{s=1}^{T} \mathsf{Var}\left(\frac{1}{\sqrt{|\mathcal{G}|_0}} \sum_{i \in \mathcal{G}} \Omega_i f_s e_{is} u_{is} \middle| F \right).$$

Before stating our final assumption, it is useful to define a number of objects.

$$\begin{aligned} b_{NT,1} &= \max_{t \leq T} \| \frac{1}{NT} \sum_{i=1}^{N} \sum_{s=1}^{T} w_s (e_{is} e_{it} - \mathsf{E} e_{is} e_{it}) \| \\ b_{NT,2} &= \left(\max_{t \leq T} \frac{1}{T} \sum_{s=1}^{T} (\frac{1}{N} \sum_{i=1}^{N} e_{is} e_{it} - \mathsf{E} e_{is} e_{it})^2 \right)^{1/2} \\ b_{NT,3} &= \max_{t \leq T} \| \frac{1}{N} \sum_{i=1}^{N} l_i e_{it} \|, \qquad b_{NT,4} = \max_{i \leq N} \| \frac{1}{T} \sum_{s=1}^{T} e_{is} w_s \| \\ b_{NT,5} &= \max_{i \leq N} \| \frac{1}{NT} \sum_{j=1}^{N} \sum_{s=1}^{T} l_j (e_{js} e_{is} - \mathsf{E} e_{js} e_{is}) \|. \end{aligned}$$

In addition, we introduce Hessian matrices that appear in the iterative estimation of λ_i and f_t.

$$\begin{aligned} D_{ft} &= \frac{1}{N} \Lambda' (\mathsf{diag}(X_t) M_\alpha \mathsf{diag}(X_t) \Lambda, \qquad D_{\lambda i} = \frac{1}{T} F' (\mathsf{diag}(\underline{X}_i) M_g \mathsf{diag}(\underline{X}_i)) F, \\ \bar{D}_{ft} &= \frac{1}{N} \Lambda' \mathsf{E}((\mathsf{diag}(e_t) M_\alpha \mathsf{diag}(e_t)) \Lambda + \frac{1}{N} \Lambda' (\mathsf{diag}(L w_t) M_\alpha \mathsf{diag}(L w_t) \Lambda, \\ \bar{D}_{\lambda i} &= \frac{1}{T} F' \mathsf{E}(\mathsf{diag}(E_i) M_g \mathsf{diag}(E_i)) F + \frac{1}{T} F' (\mathsf{diag}(W l_i) M_g \mathsf{diag}(W l_i)) F. \end{aligned}$$

The above matrices involve a number of new objects. X_t and e_t respectively denote the $N \times 1$ vectors obtained by stacking x_{it} and e_{it} for a given t. Similarly, $\underline{X_i}$ and E_i respectively denote the $T \times 1$ vectors obtained by stacking x_{it} and e_{it} for a given i. L denotes the

$N \times \dim(w_t)$ matrix composed of l_i, and W denotes the $T \times \dim(w_t)$ matrix composed of the w_t. $M_g = I - G(G'G)^{-1}G'$ is a $T \times T$ projection matrix, and M_α is an $N \times N$ matrix defined similarly. Finally, $\mathsf{diag}(e_t)$ denote the diagonal matrix whose entries are elements of e_t, and all other $\mathsf{diag}(.)$ matrices are defined similarly.

Finally, let $C_{NT} := \min\{\sqrt{N}, \sqrt{T}\}$.

Assumption 3.5 (Moment bounds). *(i)* $\max_i(\|\lambda_i\| + \|\alpha_i\| + \|l_i\|) < C$.

(ii) $\max_{t\leq T} \|\frac{1}{N}\sum_i e_{it}\alpha_i\lambda_i'\|_F = o_P(1)$ *and* $\delta_{NT}\max_{it}|e_{it}| = o_P(1)$, *where*

$$\delta_{NT} := (C_{NT}^{-1} + b_{NT,4} + b_{NT,5})\max_{t\leq T}\|w_t\| + b_{NT,1} + b_{NT,3} + C_{NT}^{-1}b_{NT,2} + C_{NT}^{-1/2}.$$

(iii) Let $\psi_j(H)$ denote the j^{th} largest singular value of matrix H. Suppose there is $c > 0$, so that almost surely, for all $t \leq T$ and $i \leq N$, $\min_{j\leq K_2}\psi_j(D_{\lambda i}) > c$, $\min_{j\leq K_2}\psi_j(D_{ft}) > c$, $\min_{j\leq K_2}\psi_j(\bar{D}_{\lambda i}) > c$ *and* $\min_{j\leq K_2}\psi_j(\bar{D}_{ft}) > c$. *In addition,*

$$c < \min_j \psi_j\left(\frac{1}{N}\sum_i l_i l_i'\right) \leq \max_j \psi_j\left(\frac{1}{N}\sum_i l_i l_i'\right) < C,$$
$$c < \min_j \psi_j\left(\frac{1}{T}\sum_t w_t w_t'\right) \leq \max_j \psi_j\left(\frac{1}{T}\sum_t w_t w_t'\right) < C.$$

(iv) $\max_{it}\mathsf{E}(e_{it}^8|U, F) < C$, $\mathsf{E}\|w_t\|^4 + \mathsf{E}\|g_t\|^4 + \mathsf{E}\|f_t\|^4 < C$, *and*

$$\mathsf{E}\|g_t\|^4\|f_t\|^4 + \mathsf{E}u_{it}^4\|f_t\|^4 + \mathsf{E}e_{jt}^4\|f_t\|^8 + \mathsf{E}e_{jt}^4\|f_t\|^4\|g_t\|^4 + \mathsf{E}\|w_t\|^4\|g_t\|^4 < C.$$

Assumption 3.5 (i) imposes that factor loadings, which are treated as non-random, are bounded. Assumption 3.5 (ii) then imposes tail conditions on (e_{it}, w_t). If (e_{it}, w_t) are sub-Gaussian and e_{it} is independent across i, it is straightforward to obtain bounds for $b_{NT,1}, ..., b_{NT,5}$ under which one can verify that condition (ii) holds as long as $\log^2 T = o(N)$ and $(\log^2 T)(\log^2 N) = o(T)$. Condition (iii) imposes relatively mild restrictions on Hessian and loading matrices that guarantee strong identification of the model parameters. Finally, Assumption 3.5 (iv) imposes relatively standard moment bounds on the factors.

3.3 Main Results

Assume that $\mathcal{G}$ is a known cross-sectional subset of interest. Theorem 3.1 verifies that cross-sectional averages of the heterogeneous effects are asymptotically normal.

Theorem 3.1. *Suppose Assumptions 3.1-3.5 hold. Fix any $t \leq T$ and $\mathcal{G} \subseteq \{1, ..., N\}$. Suppose $N, T \to \infty$ and either (i) $|\mathcal{G}|_0 = o(\min\{N/T, T\})$ or (ii) $N = O(T|\mathcal{G}|_0)$ with $N = o(T^2)$. In addition, assume $f_t' V_f f_t$ and $\bar{\lambda}_{\mathcal{G}}' V_\lambda \bar{\lambda}_{\mathcal{G}}$ are both bounded away from zero. Then*

$$\Sigma_{\mathcal{G}}^{-1/2} \left(\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \widehat{\theta}_{it} - \bar{\theta}_{\mathcal{G},t} \right) \to^d \mathcal{N}(0,1)$$

where

$$\Sigma_{\mathcal{G}} := \frac{1}{T|\mathcal{G}|_0} f_t' V_{f,\mathcal{G}} f_t + \frac{1}{N} \bar{\lambda}_{\mathcal{G}}' V_\lambda \bar{\lambda}_{\mathcal{G}}$$

with

$$V_\lambda \quad = \quad V_{\lambda 1}^{-1} V_{\lambda 2} V_{\lambda 1}^{-1}, \quad V_{\lambda 1} = \frac{1}{N} \sum_{i=1}^N \lambda_i \lambda_i' \mathsf{E} e_{it}^2, \quad \bar{\lambda}_{\mathcal{G}} = \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \lambda_i.$$

Remark 3.1. Theorem 3.1 implies that the group average effect estimator has rate of convergence[2]

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \widehat{\theta}_{it} - \bar{\theta}_{\mathcal{G},t} = O_P \left(\frac{1}{\sqrt{T|\mathcal{G}|_0}} + \frac{1}{\sqrt{N}} \right).$$

For a fixed group of interest $\mathcal{G}$, it is useful to compare this rate to the rate that would be obtained using a model that imposes homogeneous effects at the group level:

$$y_{it} = \theta_{\mathcal{G},t} x_{it} + \alpha_i' g_t + u_{it}, \quad i \in \mathcal{G}. \tag{3.1}$$

[2] The rate of convergence can be obtained under weaker conditions than needed for the distributional approximation in Theorem 3.1.

The resulting "group-homogeneous effect estimator" $\widetilde{\theta}_{G,t}$ would satisfy

$$\widetilde{\theta}_{G,t} - \theta_{\mathcal{G},t} = O_P\left(|\mathcal{G}|_0^{-1/2}\right).$$

In effect, the homogeneous effect estimator uses only cross-sectional information within $\mathcal{G}$. By leveraging the factor coefficients, we estimate a heterogeneous average effect $\bar{\theta}_{\mathcal{G},t} = \frac{1}{|\mathcal{G}|_0}\sum_{i\in\mathcal{G}}\theta_{it}$ making use of all cross-sectional information. As a result, the estimator under the factor coefficient structure will have a faster rate of convergence than $\widetilde{\theta}_{G,t}$ when $|\mathcal{G}|_0$ is small and will remain consistent even if the group size is finite. We also note that the homogeneous effects estimator will generally be inconsistent for group average effects when effects are actually heterogeneous. It is important to note that we are estimating the average effect for a given *known* group. See, e.g., Bonhomme and Manresa (2015) and Su et al. (2016) for results on estimating group effects when group memberships are also estimated from the data.

Remark 3.2. Suppose $\mathcal{G} = \{i\}$, i.e., suppose we aim to estimate the effect for a specific individual i. In this case, the required condition on the growth of $(|\mathcal{G}|_0, N, T)$—that either (i) $1 = o(\min\{N/T, T\})$ or (ii) $N = O(T)$ holds—is *always* satisfied regardless of the relative growth rate between N and T (provided that both grow to infinity). This result matches classical results for pure factor models, such as Theorem 3 in Bai (2003), where inference about $\lambda_i' f_t$ does not impose any condition on the relative growth rate between N and T.

Of course, using Theorem 3.1 for inference requires that we estimate the asymptotic variance. To preserve the rotation invariance property of the asymptotic variance, we estimate relevant quantities separately within subsamples and produce the final asymptotic variance estimator by averaging the results. Specifically, assuming products $e_{it}u_{it}$ are cross-sectionally independent for simplicity, let

$$\widehat{v}_\lambda \;=\; \frac{1}{2}(\widehat{\lambda}_{I,\mathcal{G}}'\widehat{V}_{\lambda 1,I}^{-1}\widehat{V}_{\lambda 2,I}\widehat{V}_{\lambda 1,I}^{-1}\widehat{\lambda}_{I,\mathcal{G}} + \widehat{\lambda}_{I^c,\mathcal{G}}'\widehat{V}_{\lambda 1,I^c}^{-1}\widehat{V}_{\lambda 2,I^c}\widehat{V}_{\lambda 1,I^c}^{-1}\widehat{\lambda}_{I^c,\mathcal{G}}),$$

$$
\begin{aligned}
\widehat{v}_f &= \frac{1}{2}(\widehat{f}'_{I,t}\widehat{V}_{I,f}\widehat{f}_{I,t} + \widehat{f}'_{I^c,t}\widehat{V}_{I^c,f}\widehat{f}_{I^c,t}),\\
\widehat{V}_{S,f} &= \frac{1}{|\mathcal{G}|_0|S|_0}\sum_{s\notin S}\sum_{i\in\mathcal{G}}\widehat{\Omega}_{S,i}\widehat{f}_{S,s}\widehat{f}'_{S,s}\widehat{\Omega}_{S,i}\widehat{e}^2_{is}\widehat{u}^2_{is}, \quad S\in\{I,I^c\}\\
\widehat{V}_{\lambda 1,S} &= \frac{1}{N}\sum_j \widehat{\lambda}_{S,j}\widehat{\lambda}'_{S,j}\widehat{e}^2_{jt}, \quad \widehat{V}_{\lambda 2,S} = \frac{1}{N}\sum_j \widehat{\lambda}_{S,j}\widehat{\lambda}'_{S,j}\widehat{e}^2_{jt}\widehat{u}^2_{jt},\\
\widehat{\lambda}_{S,\mathcal{G}} &= \frac{1}{|\mathcal{G}|_0}\sum_{i\in\mathcal{G}}\widehat{\lambda}_{S,i}, \text{ and } \widehat{\Omega}_{S,i} = \left(\frac{1}{|S|_0}\sum_{s\in S}\widehat{f}_{S,s}\widehat{f}'_{S,s}\right)^{-1}\left(\frac{1}{T}\sum_{s=1}^T \widehat{e}^2_{is}\right)^{-1}.
\end{aligned}
$$

$\Sigma_{\mathcal{G}}$ can then be consistently estimated by

$$
\widehat{\Sigma}_{\mathcal{G}} := \frac{\widehat{v}_f}{T|\mathcal{G}|_0} + \frac{\widehat{v}_\lambda}{N}.
$$

3.4 Policy relevant tests

We now show how we can use our results to test the hypotheses described in Section 3.1. As part of this discussion, we illustrate leveraging the structure of the coefficients in the factor-slope model, $\bar{\theta}_{\mathcal{G},t} = \bar{\lambda}'_{\mathcal{G}}f_t$, to provide tractable tests.

Consider testing the null hypothesis of homogeneity of group average effects in a given time period t:

$$
H_0^1 : \bar{\theta}_{\mathcal{G}_1,t} = ... = \bar{\theta}_{\mathcal{G}_J,t}.
$$

Let $S := (\widehat{\theta}_{\mathcal{G}_1,t} - \widehat{\theta}_{\mathcal{G}_2,t}, \widehat{\theta}_{\mathcal{G}_2,t} - \widehat{\theta}_{\mathcal{G}_3,t}, ..., \widehat{\theta}_{\mathcal{G}_{J-1},t} - \widehat{\theta}_{\mathcal{G}_J,t})'$. We define the test statistic

$$
TS'(\Xi\widehat{D}\Xi')^{-1}S \text{ where } \Xi = \begin{pmatrix} 1 & -1 & 0 & \cdots & \\ 0 & 1 & -1 & 0 & \cdots \\ \vdots & & & & \\ 0 & \cdots & & 1 & -1 \end{pmatrix}
$$

and $\widehat{D} = \mathsf{diag}\{\frac{1}{|\mathcal{G}_j|_0}\widehat{v}_{f,\mathcal{G}_j} : j \le J\}$ is the diagonal matrix of the estimated asymptotic variances of $\widehat{\theta}_{\mathcal{G}_j,t}$. It is interesting to note that this test is a test of group homogeneity of loadings in the sense that $\bar{\theta}_{\mathcal{G}_j,t} - \bar{\theta}_{\mathcal{G}_k,t} = (\bar{\lambda}_{\mathcal{G}_j} - \bar{\lambda}_{\mathcal{G}_k})'f_t$ for any two groups $\mathcal{G}_j, \mathcal{G}_k$. Therefore, H_0^1 is the same as the null hypothesis of homogeneity of group average loadings except when the inner product between $\bar{\lambda}_{\mathcal{G}_j} - \bar{\lambda}_{\mathcal{G}_k}$ and f_t happens to be zero.

Next, consider the test of joint significance at a given time t:

$$H_0^2 : \theta_{it} = 0 \text{ for all } i.$$

The factor slope structure brings us at least two benefits to implement this test. First, although this hypothesis is high-dimensional, the problem can be transformed to testing a much simpler low-dimensional hypothesis: $H_0^{2\prime} : f_t = 0$. Under H_0^2, the factor structure yields $\Lambda f_t = 0$ where Λ is the $N \times \dim(f_t)$ matrix of λ_i. Suppose $(\Lambda'\Lambda)^{-1}$ exists, then H_0^2 is equivalent to $f_t = 0$ almost surely at the given time t of interest, which follows from left multiplication by $(\Lambda'\Lambda)^{-1}\Lambda'$. A secondary benefit of focusing on f_t is that failing to reject $f_t = 0$ will also suggest one would not reject $\theta_{it} = 0$ for any out-of-sample cross-sectional unit i influenced by the same factors.

Focusing on $H_0^{2\prime}$, a simple test can be constructed from

$$\widehat{f}_t := \frac{1}{2}(\widehat{f}_{I,t} + \widehat{f}_{I^c,t})$$

We can show that, when $N = o(T^2)$, there is a rotation matrix H_1 such that

$$\sqrt{N}(H_1^{-1}V_\lambda H_1'^{-1})^{-1/2}(\widehat{f}_t - H_1^{-1}f_t) \to^d \mathcal{N}(0, I).$$

Further, $H_1^{-1}V_\lambda H_1'^{-1}$ can be consistently estimated by

$$\widehat{V}_\lambda := \frac{1}{2}\widehat{V}_{\lambda 1,I}^{-1}\widehat{V}_{\lambda 2,I}\widehat{V}_{\lambda 1,I}^{-1} + \frac{1}{2}\widehat{V}_{\lambda 1,I^c}^{-1}\widehat{V}_{\lambda 2,I^c}\widehat{V}_{\lambda 1,I^c}^{-1}.$$

The following theorem presents the asymptotic null distribution of the two tests. Let $\chi^2(a)$ denote the centered chi-square distribution with degrees of freedom a.

Theorem 3.2. *(i) For H_0^1, suppose $\bar{\lambda}_{\mathcal{G}_1} = ... = \bar{\lambda}_{\mathcal{G}_J}$ under H_0^1. Assume $\max_{j\leq J}|\mathcal{G}_j|_0 = o(T)$, $T\max_{j\leq J}|\mathcal{G}_j|_0 = o(N^2)$, $\max_j |\mathcal{G}_j|_0/\min_j |\mathcal{G}_j|_0 = O(1)$ and J is fixed. Under H_0^1,*

$$TS'(\Xi\widehat{D}\Xi')^{-1}S \to^d \chi^2(J-1).$$

(ii) For H_0^2 at a fixed t of interest, suppose $f_t = 0$ almost surely under H_0^2. Then under H_0^2,

$$N\widehat{f}_t'\widehat{V}_\lambda^{-1}\widehat{f}_t \to^d \chi^2(\dim(f_t)).$$

4 Monte Carlo Simulations

In this section, we report results from a set of simulation exercises. In all cases, we specify the outcome model as

$$y_{it} = \alpha_i g_t + x_{it,1}\theta_{it} + x_{it,2}\beta_{it} + u_{it}$$

with $\dim(\alpha_i) = \dim(g_t) = 1$. We generate observed regressors from the model

$$x_{it,r} = l_{i,r}'w_{t,r} + \mu_x + e_{it,r}, \quad r = 1, 2$$

with $\mu_x = 2$, so $x_{it,r}$ follows a factor model with an intercept. We generate $\{e_{it,r}\}_{r=1,t=1}^{2,T}$ and $\{u_{it}\}_{t=1}^T$ from mutually independent AR(1) models with Gaussian innovations and autocorrelation coefficient ρ. The AR(1) models initialize from zero and then proceed as

$$u_{it} = \rho u_{i,t-1} + \mathcal{N}(0, \sigma_i^2)$$

where $\sigma_i^2 \sim \text{Uniform}[0.5, 1.5]$ independently for each unit $i = 1, ..., N$. $e_{it,r}$ is generated similarly. Variances are regenerated in each simulation replication and produce heteroskedasticity. We consider $\rho \in \{0, .5\}$, where the $\rho = .5$ case provides some evidence about the performance of our method under a violation of the intertemporal independence assumption. We generate bounded loadings for the interactive fixed effect:

$$\alpha_i = \text{sgn}(a_i) \min\{|a_i|, 10\} + 2, \quad a_i \sim \mathcal{N}(2, 1), \quad g_t \sim \mathcal{N}(2, 1).$$

To generate the heterogeneous slopes, we generate i.i.d. "slope factors" $(f_{t,1}, f_{t,2})$ and "slope loadings" $(\lambda_{i,1}, \lambda_{i,2})$, with $\dim(\lambda_{i,k}) = \dim(f_{t,k}) = 2$ for $k = 1, 2$. We generate the factors $(f_{t,1}, f_{t,2})$ as i.i.d draws from $\mathcal{N}(2, 1)$ random variables, and we generate loadings $(\lambda_{i,1}, \lambda_{i,2})$ by taking i.i.d. draws from $\mathcal{N}(2, 1)$ random variables and then setting any realization outside of ± 10 to ± 10 respectively. Letting $\bar{\lambda}_k = \frac{1}{N} \sum_{i=1}^{N} \lambda_{i,k}$ and $\bar{f}_k = \frac{1}{T} \sum_{t=1}^{T} f_{t,k}$, we then consider five designs covering different scenarios for slope heterogeneity:

- **Design I. "(i, t)-heter":** $\theta_{it} = \lambda_{i,1}' f_{t,1}$ and $\beta_{it} = \lambda_{i,2}' f_{t,2}$. This design exhibits general low-rank heterogeneous coefficients.

- **Design II. "i-heter":** $\theta_{it} = \lambda_{i,1}' \bar{f}_1$ and $\beta_{it} = \lambda_{i,2}' \bar{f}_2$. This design corresponds to a model with intertemporal homogeneity but unit-level heterogeneity in coefficients as studied, e.g., in Pesaran (2006).

- **Design III. "t-heter":** $\theta_{it} = \bar{\lambda}_1' f_{t,1}$ and $\beta_{it} = \bar{\lambda}_2' f_{t,2}$. This design corresponds to a model with cross-sectional homogeneity but time-varying heterogeneity in coefficients.

- **Design IV. "homo":** We set $\theta_{it} = \bar{\lambda}_1' \bar{f}_1$ and $\beta_{it} = \bar{\lambda}_2' \bar{f}_2$. This design corresponds to a model with homogeneous (constant) coefficients as studied, e.g., in Bai (2009).

- **Design V. "2-way":** $\theta_{it} = \lambda_{i,1}' f_{t,1}$ and $\beta_{it} = \lambda_{i,2}' f_{t,2}$ (exactly as in Design I), but $\alpha_i g_t$ is replaced by $\alpha_i + g_t$. That is, this design exhibits general low-rank heterogeneous effects but specifies the "intercept" to correspond to the additive two-way-fixed-effects structure.

We note that all of these designs are special cases of our general low-rank heterogeneous effects model. We apply our proposed low-rank estimation procedure within each design using the tuning parameter choices and data-driven method for choosing the rank of the slope coefficients described in Section 2.4. For simplicity, we report results focusing on $\theta_{1,1}$. We first report coverage obtained by applying the distributional result and standard error estimator from Section 3.3. We then report comparisons to the estimator that directly solves (2.3) using the tuning parameter choices outlined in Section 2.4 but without sample-splitting or the subsequent debiasing steps and to other estimators that specifically impose the structure of Design II-Design V. We conclude by looking at using our method to test the hypothesis of homogeneity of average effects across a set of pre-specified groups.

4.1 Coverage and Distributional Approximation

Table 6.1 reports simulation coverage probabilities for $\theta_{1,1}$ based on 1,000 simulation replications across our five designs. We consider two different sample size configurations: $(N, T) = (50, 50)$ and $(N, T) = (500, 100)$. The first two columns ($\rho = 0$) provide coverage in the case where innovations are temporally independent, while the last two columns ($\rho = 0.5$) report results in the case with temporal dependence.

Across all cases, we see that the proposed procedure performs relatively well. We see mild size distortions in the smaller sample sizes that disappear in the independent case in the larger sample. As the formal theory does not allow for dependence, it is unsurprising that we see size distortions in the dependent case that seem to persist as the sample size increases. Though certainly depending on specific DGP settings, it is somewhat reassuring that these distortions appear small, at least in this case.

Next, we assess the role of the iterative least squares steps in improving inference beyond the initial nuclear-norm penalization. Figure 6.1 shows the (scaled) histogram of the z-values for the estimated $\theta_{1,1}$ across all designs, overlaid with the standard normal density in the $N = 50$, $T = 50$ case.[3] We also compare our proposed method to a "regularized" approach that relies solely on nuclear-norm penalization. In this alternative, $\theta_{1,1}$ is estimated directly

[3] An analogous figure for the case with $N = 500$ and $T = 100$ is provided in the supplementary appendix.

as the $(1,1)$ entry of $\widetilde{\Theta}$ from equation (2.3). Since the "regularized" method lacks a valid asymptotic theory, its z-values are standardized using the (infeasible) simulation standard deviations.

Across all designs and both $\rho = 0$ and $\rho = 0.5$, the z-values from our proposed estimator closely follow a symmetric distribution that aligns well with the standard normal. In contrast, the distribution of z-values under the "regularized" method exhibits clear bias. These findings suggest that the iterative least squares steps primarily enhance inference by correcting the shrinkage bias introduced by regularization.

4.2 Bias and Efficiency

We now consider point estimation properties of our proposed estimator in comparison to several other approaches that can be employed to estimate slope coefficients in panel data. Specifically, we compare the following estimators:

- **"ours":** Our proposed estimator.
- **"regul":** The nuclear-norm regularized estimator solving (2.3).
- **"*i*-heter":** The estimator that solves $\min_{\alpha_i, g_t, \beta_i} \sum_{i,t} (y_{it} - \alpha_i g_t - x_{it}'\beta_i)^2$ — that is, the estimator obtained by imposing time homogeneity of coefficients.
- **"*t*-heter":** The estimator that solves $\min_{\alpha_i, g_t, \beta_t} \sum_{i,t} (y_{it} - \alpha_i g_t - x_{it}'\beta_t)^2$ — that is, the estimator obtained by imposing homogeneity of coefficients across units.
- **"homo":** The estimator that solves $\min_{\alpha_i, g_t, \beta} \sum_{i,t} (y_{it} - \alpha_i g_t - x_{it}'\beta)^2$ — that is, the estimator obtained by imposing full homogeneity of coefficients from Bai (2009).

We report simulation absolute value of bias (multiplied by 10) and mean squared error (MSE) from applying each of these estimators in each of our five designs in Table 6.2. The first four columns contain results from the design with temporally independent innovations, while the last four columns report results in the dependent case. All results are obtained using 1000 simulation replications.

The results across the independent and dependent cases are broadly consistent with each other. Our method is designed for a model with general heterogeneous slopes which includes homogeneous slopes as a special case. As such, it is not surprising to see that it performs relatively well across all considered designs. Relative to the simple regularized estimator that does not employ sample-splitting or debiasing, we see that our approach has significantly less bias while maintaining comparable performance in terms of simulation MSE. This performance mirrors the results from the previous section. All competing methods treat the fixed effect as interactive (i.e., $\alpha_i' g_t$), which reduces the estimation efficiency when the true effects are in fact 2-way additive as in Design V. This explains why the estimation is slightly worse in Design V than in Design I.

We do see that, within the homogeneous designs, our procedure performs somewhat less well than the estimators that are specifically designed to exploit the homogeneity. However, these estimators perform much worse than our proposed procedure when the design has heterogeneity that does not align with the type of homogeneity imposed by the specific estimator. Of course, researchers typically do not know the true structure of heterogeneity in the data in practice. Thus, the strong and stable performance of our method across a variety of designs is a particularly attractive feature.

5 Empirical Illustration: Estimating the Effect of the Minimum Wage on Employment

5.1 Background

To illustrate our approach, we estimate the effect of the minimum wage on youth employment. Previous studies in the literature reach mixed conclusions. For instance, Card and

Krueger (1994) find that the minimum wage has a positive effect on employment using data of restaurants in New Jersey and Eastern Pennsylvania. Dube et al. (2010) find negative employment effects when controlling for county-level population but no adverse effects when identification relies solely on variation among contiguous county pairs. Wang et al. (2019) allow for slope heterogeneity at the county level by estimating the model

$$y_{it} = x_{it,1}\theta_i + x_{it,2}\beta_i + \alpha_i + g_t + u_{it} \tag{5.1}$$

where i denotes county, t indexes quarter, y_{it} is log employment, $x_{it,1}$ is the log minimum wage, $x_{it,2}$ is log population, and α_i and g_t are respectively additive county and time fixed effects. They estimate county-level heterogeneous effects by imposing a grouping structure where θ_i and β_i take on a small number of distinct values that are homogeneous within groups. The values of the group effects and group membership of each county is then learned as part of their estimation algorithm. Their findings suggest mixed effects of the minimum wage across groups in the sense that they find groups where estimated effects are positive and groups where estimated effects are negative. See also Neumark and Wascher (2008) for an overview and more recent work by Cengiz et al. (2019) and Callaway and Sant'Anna (2021).

We contribute to this literature by estimating our model, which allows for a rich intertemporal and cross-sectional heterogeneity structure in both the minimum wage effect and the fixed effects structure. It seems interesting to allow for intertemporal and cross-sectional heterogeneity of effects as there seems to be no *ex ante* reason to believe that these effects are constant. For example, these effects are presumably related to pressures from both the supply and demand side and other market conditions, all of which are plausibly time varying and related to local labor market conditions. Allowing for the interactive fixed effects structure may also be useful for capturing potentially complex unobserved heterogeneity that is correlated to the minimum wage.

5.2 Implementation

We use state-level data analogous to the county-level data in Dube et al. (2010) and Wang et al. (2019).[4] The data are a balanced panel with $T = 66$ and $N = 51$ observations on the 50 states in the United States plus Washington D.C. running from the first quarter of 1990 to the second quarter of 2006. Using these data, we estimate a model for log youth employment (y_{it}):

$$y_{it} = x_{it,1}\theta_{it} + x_{it,2}\beta_{it} + x_{it,3}\gamma_{it} + \alpha_i' g_t + u_{it}$$

with

$$\theta_{it} = \lambda_{i,1}' f_{t,1}, \quad \beta_{it} = \lambda_{i,2}' f_{t,2}, \quad \gamma_{it} = \lambda_{i,3}' f_{t,3},$$

where $x_{it,1}$, $x_{it,2}$, and $x_{it,3}$ respectively denote the log minimum wage, log state population, and log total employment.

For the orthogonalization step of our estimation procedure, we model each regressor using a factor structure:

$$x_{it,k} = \mu_{it,k} + e_{it,k}, \quad k = 1, 2, 3.$$

where $\mu_{it,k} = \alpha_{ik} + l_{i,k}' w_{t,k}$. We treat α_{ik} as additive fixed effects and allow for the presence of additional factors by including $l_{i,k}' w_{t,k}$. This structure ensures that there is at least one factor in every variable - a factor that is always one. The remaining factor structure can then pick up any other sources of strong correlation and is expected to pick up the general trend that all of the variables we consider tend to increase over the sample period.

We apply the eigenvalue-ratio method of Ahn and Horenstein (2013) to each regressor. This approach estimates that there are three factors in each regressor. To estimate the $\mu_{it,k}$,

[4] We are grateful to Wuyi Wang for sharing the data with us.

we thus employ PCA with three factors on the demeaned data for each regressor:

$$\dot{x}_{it,k} = x_{it,k} - \frac{1}{T}\sum_{t=1}^{T} x_{it,k} - \frac{1}{N}\sum_{i=1}^{N} x_{it,k} + \frac{1}{TN}\sum_{t=1}^{T}\sum_{i=1}^{N} x_{it,k}$$

We then use the resulting estimates of $\mu_{it,k}$ in Algorithm 2.2 in order to estimate the heterogeneous effects θ_{it}.

5.3 Results

We estimate the minimum wage effect for each state and quarter. To summarize the results, we group the states into two equal-size groups according to their average minimum wage over the sample period: high minimum wage states (High) and low minimum wage states (Low).

The left panel of Figure 6.2 plots the averaged effects $\bar{\theta}_{\mathcal{G},t}$ for $\mathcal{G} \in \{\text{Low, High}\}$ across time. The most striking feature of the graph is that the estimated average minimum wage effect within the low minimum wage group is much more volatile than the estimated average effect within the high minimum wage states. This difference is especially pronounced in the early part of the sample period where we see extreme swings between estimated effects. In the latter part of the sample period, the estimated average effects in both the low and high minimum wage groups appear relatively stable and concentrated on relatively small values. We also see that estimated effects in the low-minimum wage group are associated with much larger standard errors than effects estimated in the high-minimum wage group. It is interesting that broadly speaking, the estimated effects seem to follow the same pattern in both groups with estimated positive and negative effects in both groups tending to occur in the same time periods.

For comparison, we also plot the estimated effects in two specific states — Oregon, which has a high average minimum wage over the sample period, and North Carolina, which has a low average minimum wage over the sample period — in the right panel of Figure 6.2. The depicted effects roughly follow the general patterns exhibited by the group level averages, though with somewhat more volatility. We also see that intervals are, of course, larger when looking at estimates for individual states.

We also conduct a test of homogeneity within each group:

$$H_0 : \theta_{i,t} \text{ are all the same}, \quad \forall i \in \mathcal{G}, \forall t \leq T$$

where our groups are again the high and low minimum wage states.[5] Applying this test in the high minimum wage group yields a near-zero p-value, and we obtain a p-value of 0.021 in the low minimum-wage group. These test results are consistent with the visual evidence, suggesting significant heterogeneity of minimum wage effects.

As a point of reference, we also obtained an estimate of the minimum wage effect using the usual homogeneous two-way fixed effects model:

$$y_{it} = x_{it,1}\theta + x_{it,2}\beta + x_{it,3}\gamma + \alpha_i + g_t + u_{it}.$$

Within our data, the estimate of θ is -0.1128 (s.e. $=$ 0.014), where the standard error is obtained using two-way clustering by state and time. If we estimate this model within subsets formed from the high and low minimum wage states, we obtain estimates of -0.047 (s.e. $=$ 0.014) within the high minimum wage states and -0.304 (s.e. $=$ 0.030) within the low minimum wage states. While these results are significant, they ignore the time-heterogeneity and thus the drawn conclusions differ sharply from those one would draw using the richer heterogeneous effects model where we find substantial evidence for heterogeneous effects over time.

An interesting extension is to allow both additive and interactive fixed effects, by considering

$$y_{it} = \theta_{it}' x_{it} + \alpha_i' g_t + h_t + z_i + e_{it}$$

[5] We provide a formal treatment of testing this hypothesis in Section D.2 in the supplementary appendix.

where α_i, g_t, h_t and z_i are all fixed effects. While interactive effects technically nest additive effects, explicitly including additive effects has the potential to improve performance in finite samples and also mechanically nests the workhorse two way fixed effects model. Within this structure, estimation could proceed via penalized regression by solving

$$\min_{\Theta,M,H,Z} \|Y - \Theta \odot X - M - 1_N H' - Z 1_T'\|_F^2 + \nu_0 \|M\|_n + \nu_1 \|\Theta\|_n$$

where $H \in \mathbb{R}^T$ and $Z \in \mathbb{R}^N$. While interesting, solving this program is more complex than the one without additive effects and would require different, though similar, theory. Given that the interactive effects model technically nests this model and is easier to solve, we do not consider the model with explicit additive effects further. It does seem like exploring this model is an interesting direction for future research.

6 Conclusion

We study estimation of and inference for slope coefficients in a linear panel data model with interactive fixed effects and slopes that vary over time and across individuals. To make learning possible, we impose that these individual/time varying parameters follow a factor structure. As part of our debiasing procedure we make use of sample splitting, which could potentially be avoided if one could directly establish that these terms vanish using the full sample. In that case, we conjecture our estimator would have the same asymptotic distribution as the full sample estimator. Our proposed approach relies on having large N and large T. These conditions suggest that our approach may perform poorly in applications where either N or T is relatively small. It should be possible to relax these conditions if additional structure is available for restricting the model for the low-rank matrices.

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Table 6.1: Coverage Probability

Design	$\rho = 0$		$\rho = 0.5$	
	$N = 50$ $T = 50$	$N = 500$ $T = 100$	$N = 50$ $T = 50$	$N = 500$ $T = 100$
(i, t)-heter	0.924	0.946	0.909	0.920
i-heter	0.928	0.952	0.920	0.933
t-heter	0.928	0.964	0.926	0.930
homo	0.930	0.951	0.923	0.928
2-way	0.911	0.950	0.898	0.929

This table reports the simulated coverage probability of 95% confidence intervals obtained from our proposed method under five different designs of heterogeneity. The row label gives the type of heterogeneity allowed for in the simulation design. Results are based on 1000 simulation replications.

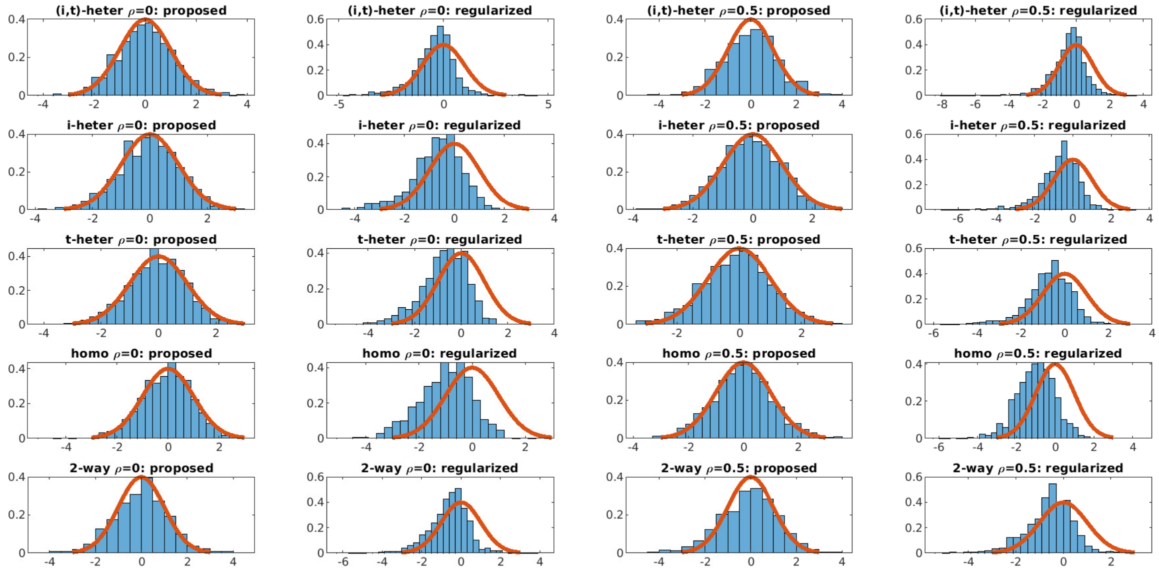

Figure 6.1: Each panel displays the histograms of a standardized estimate — $(\widehat{\theta}_{11} - \theta_{11})/se(\widehat{\theta}_{11})$ — in the $N = 50, T = 50$ case in a different design. Panels with "proposed" in the label correspond to our proposed estimator standardized by the feasible standard error estimator from Section 3.3. Panels with "regularized" in the label correspond to using the nuclear-norm regularized estimator obtained from directly solving (2.3). In this case, the figure is standardized by the (infeasible) simulation standard error. The standard normal density function is superimposed on each histogram in red.

Table 6.2: Bias and MSE

	$\rho = 0$				$\rho = 0.5$			
	$N = 50, T = 50$		$N = 500, T = 100$		$N = 50, T = 50$		$N = 500, T = 100$	
Method	10|Bias|	MSE	10|Bias|	MSE	10|Bias|	MSE	10|Bias|	MSE
Design I: (i, t)-hetero								
Ours	0.849	1.206	0.391	0.180	0.535	0.096	0.997	0.642
Regular	2.601	0.301	1.212	0.067	1.335	0.083	2.827	0.341
i-hetero	1.372	10.250	1.204	10.324	1.303	11.033	0.860	11.241
t-hetero	0.395	11.286	0.924	9.899	0.516	10.691	1.515	10.691
Homo	9.430	18.332	5.943	17.637	7.899	18.895	9.259	19.180
Design II: i-hetero								
Ours	0.862	0.539	0.441	0.097	0.528	0.055	0.872	0.259
Regular	2.120	0.117	1.345	0.042	1.281	0.042	2.630	0.158
i-hetero	0.046	0.008	0.019	0.004	0.013	0.006	0.072	0.012
t-hetero	0.870	8.511	0.319	7.689	1.176	8.180	1.401	9.033
Homo	0.658	8.070	0.525	7.445	0.315	7.957	2.262	8.340
Design III: t-hetero								
Ours	0.767	0.511	0.359	0.096	0.561	0.054	1.123	0.286
Regular	2.528	0.314	1.856	0.239	1.586	0.290	2.775	0.375
i-hetero	1.171	8.673	1.601	7.752	0.759	8.462	0.179	8.131
t-hetero	0.001	0.032	0.041	0.002	0.042	0.003	0.090	0.035
Homo	0.097	8.233	2.420	7.705	0.452	8.168	1.015	7.896
Design IV: Homo								
Ours	1.170	0.559	0.430	0.090	0.742	0.056	1.033	0.269
Regular	2.777	0.173	1.702	0.102	1.963	0.122	2.778	0.170
i-hetero	0.020	0.008	0.009	0.004	0.016	0.006	0.022	0.011
t-hetero	0.061	0.027	0.042	0.002	0.025	0.003	0.017	0.027
Homo	0.034	0.000	0.118	0.000	0.103	0.000	0.032	0.000
Design V: 2-way								
Ours	1.343	1.464	0.130	0.224	0.121	0.110	1.139	0.858
Regular	3.606	0.474	2.847	0.314	2.891	0.265	4.162	0.542
i-hetero	0.589	10.803	1.179	10.536	1.742	9.936	1.203	10.590
t-hetero	0.751	10.684	0.283	10.486	0.152	10.712	0.622	10.945
Homo	10.050	19.672	12.164	20.027	11.064	19.658	11.983	20.009

Note: All values are averaged across 1000 simulations. We present ten times the absolute of bias and the simulation mean squared error.

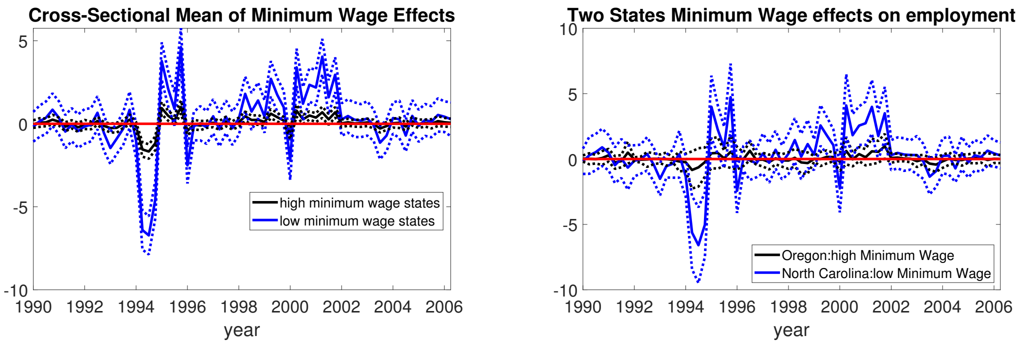

Figure 6.2: The left panel plots the time series of cross-sectional averages of estimated state-level minimum wage effects within groups. The solid black line presents the estimated average effect within the high-minimum wage group, and the solid blue curve presents the estimated average effect with the low-minimum wage group ("low MW states"). The right panel plots the time series of estimated effects within two states: Oregon (in blue) and North Carolina (in black). Dashed lines provide 95% level pointwise confidence intervals.